PHYSIOLOGY OF PREGNANCY: CLINCAL CASES AND SCENARIOS

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Thank you to Dr Belkin for sharing his slides

Learning Objectives –

After participating in this session, learners should be able to

- Describe specific physiologic changes to the cardiovascular, hematologic, pulmonary, renal, gastrointestinal, and endocrine system that occur in pregnancy

- Correlate the physiologic changes to all systems in pregnancy with the needs of the fetus and mother

- Apply the system-specific physiologic changes in pregnancy to explain normal and abnormal symptomatology in pregnancy

INTRODUCTION TO THIS LECTURE: *HEALTHY MOM HEALTHY BABY*

Understanding pregnancy physiology is critical to understanding normal and complicated pregnancies.

This lecture is an attempt to make this topic clinically relevant, interesting, and nonthreatening.

In that way, students can become comfortable with the maternal physiologic changes associated with pregnancy so that you can take good care of patients regardless of your specialty choice. This should be true even if you are not and obstetrician.



STRUCTURE OF THE LECTURE:

- My overall strategy with this lecture is to place you in the position of being an obstetrician seeing patients in a “real-world” environment.

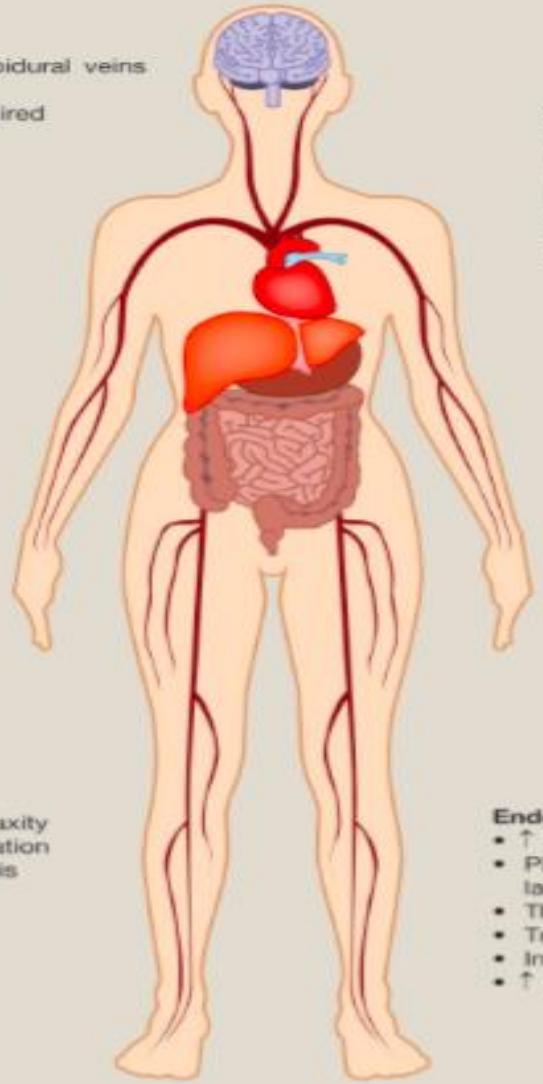
- If you have a good grasp of this material, you will be able to give solid care to your patients in whichever specialty you choose, even if it is not Obstetrics and Gynecology.

- We will use an organ-system based approach, which is how these topics are typically covered in book chapters, and test questions.

Summary of physiological changes of pregnancy

Neurological
- ↑ CSF pressure
- Engorgement of epidural veins
- ↓ MAC
- ↓ LA volumes required

Respiratory
- ↑ MV (↑TV and ↑RR)
- ↓ $PaCO_2$
- ↑ PaO_2
- ↓ FRC



Cardiac
- ↑ CO
- ↑ SV
- ↑ HR
- Left ventricular hypertrophy
- Regurgitant murmurs
- ↓ SVR

Gastrointestinal
- ↓ Lower oesophageal sphincter tone
- Reflux
- ↑ Risk of aspiration
- Liver enzymes (AST, ALT, GGT) ↓
- ↑ ALP

Renal
- ↑ Renal blood flow
- ↑ GFR
- ↓ Plasma urea and creatinine
- ↑ Urinary protein and glucose
- ↑ Risk of UTI

Musculoskeletal
- ↑ Ligamentous laxity
- ↑ Risk of dislocation
- ↑ Lumbar lordosis

Endocrine
- ↑ Progesterone and oestrogen
- Placenta secretes relaxin, human placental lactogen and human chorionic gonsdotrophin
- Thyroid hyperplasia
- Transient hyperthyroidism
- Insulin resistance
- ↑ Cortisol secretion by adrenal glands

ALP, alkaline phosphatase; ALT, alanine transaminase; AST, aspartate transaminase; CO, cardiac output; CSF, cerebrospinal fluid; FRC, functional residual capacity; GFR, glomerular filtration rate; GGT, γ-glutamyl transferase; HR, heart rate; LA, local anaesthetic; MAC, minimum alveolar concentration; MV, minute volume; SV, stroke volume; SVR, systemic vascular resistance; UTI, urinary tract infection.

CARDIOVASCULAR CHANGES IN PREGNANCY:

At a routine office visit, a 29-year-old patient reports feeling the sensation of a racing heart, and possibly heart palpitations. The patient is generally well appearing, and the blood pressure is normal at 110/60, and the heart rate is normal at 80. The patient wonders if everything is fine, or if there is some sort of problem.

PROGESTERONE AND THE NITRIC OXIDE PATHWAY:

- BUT FIRST…the progesterone and nitric oxide pathways are the key cellular mediators of the pregnancy process.
- Nuclear and novel membrane progesterone receptors are responsible for regulation of cAMP, and increased nitric oxide production, which results in increased cyclic GMP (cGMP). This ultimately results in systemic vasodilation. The exact mechanisms underlying this phenomenon remain unclear.
- Ultimately, both systems work in conjunction with each other to cause smooth muscle relaxation and uterine quiescence.

Normal cardiac changes that mimic disease:

- Some of the physiologic changes of pregnancy can seem like heart disease/heart problems (or at least be mildly concerning) to patients.

  Patients will very often complain of:
  - Dyspnea ON EXERTION
  - Pre-syncope
  - Decreased exercise tolerance
  - Palpitations

CARDIOVASCULAR CHANGES: AN EXPLANATION:

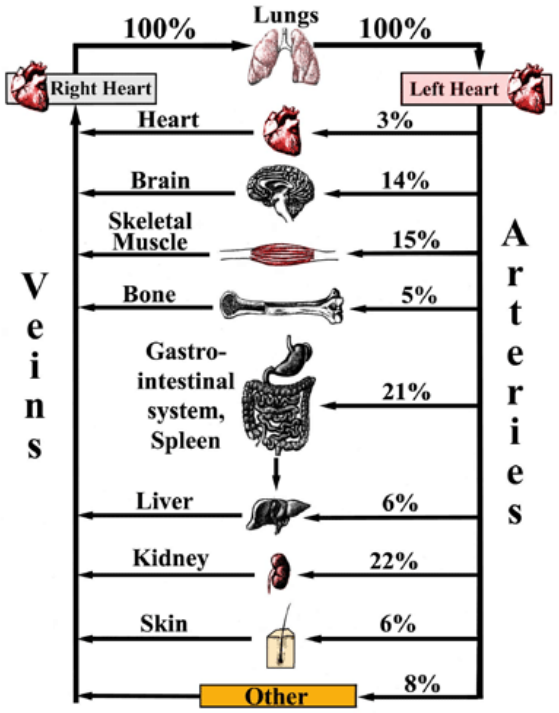


Table 1. Cardiovascular Changes in a Normal Pregnancy*

	First Trimester	Second Trimester	Third Trimester	Stage 1 Labor	Stage 2 Labor	Early Postpartum	3–6 months Postpartum
Cardiac output	↑5–10%	↑↑35–45%		↑30%	↑↑50%	↑↑↑60–80% immediately, then rapidly decreases within the first hour	Return to prepregnancy values
Heart rate	↑3–5%	↑10–15%	↑15–20%	During uterine contractions: ↑40–50%		↓5–10% within 24 hours; continues to decrease throughout the first 6 weeks	Return to prepregnancy values
Blood pressure	↓10%	↓5%	↑5%	During uterine contractions: ↑SBP 15–25% ↑DBP 10–15%		↓SBP 5–10% within 48 hours; may increase again between days 3–6 due to fluid shifts	Return to prepregnancy values
Plasma volume	↑	↑↑40–50%		↑	↑↑	↑↑↑500 mL due to autotransfusion	Return to prepregnancy values

Abbreviations: SBP, systolic blood pressure; DBP, diastolic blood pressure.

*Hemodynamic changes that occur during pregnancy, labor, and postpartum (compared with prepregnancy) should be understood to identify early interventions (such as blood pressure control and diuresis) that may be needed to prevent clinical deterioration in a woman with cardiovascular disease.

Data from Kuhn JC, Falk RS, Langesaeter E. Haemodynamic changes during labour: continuous minimally invasive monitoring in 20 healthy parturients. Int J Obstet Anesth 2017;31:74–83; Ouzounian JG, Elkayam U. Physiologic changes during normal pregnancy and delivery. Cardiol Clin 2012;30:317–29; Sanghavi M, Rutherford JD. Cardiovascular physiology of pregnancy. Circulation 2014;130:1003–8; Shen M, Tan H, Zhou S, Smith GN, Walker MC, Wen SW. Trajectory of blood pressure change during pregnancy and the role of pre-gravid blood pressure: a functional data analysis approach. Sci Rep 2017;7:6227; Sohnchen N, Melzer K, Tejada BM, Jastrow-Meyer N, Othenin-Girard V, Irion O, et al. Maternal heart rate changes during labour. Eur J Obstet Gynecol Reprod Biol 2011;158:173–8; and Walters BN, Walters T. Hypertension in the puerperium [letter]. Lancet 1987;2:330.

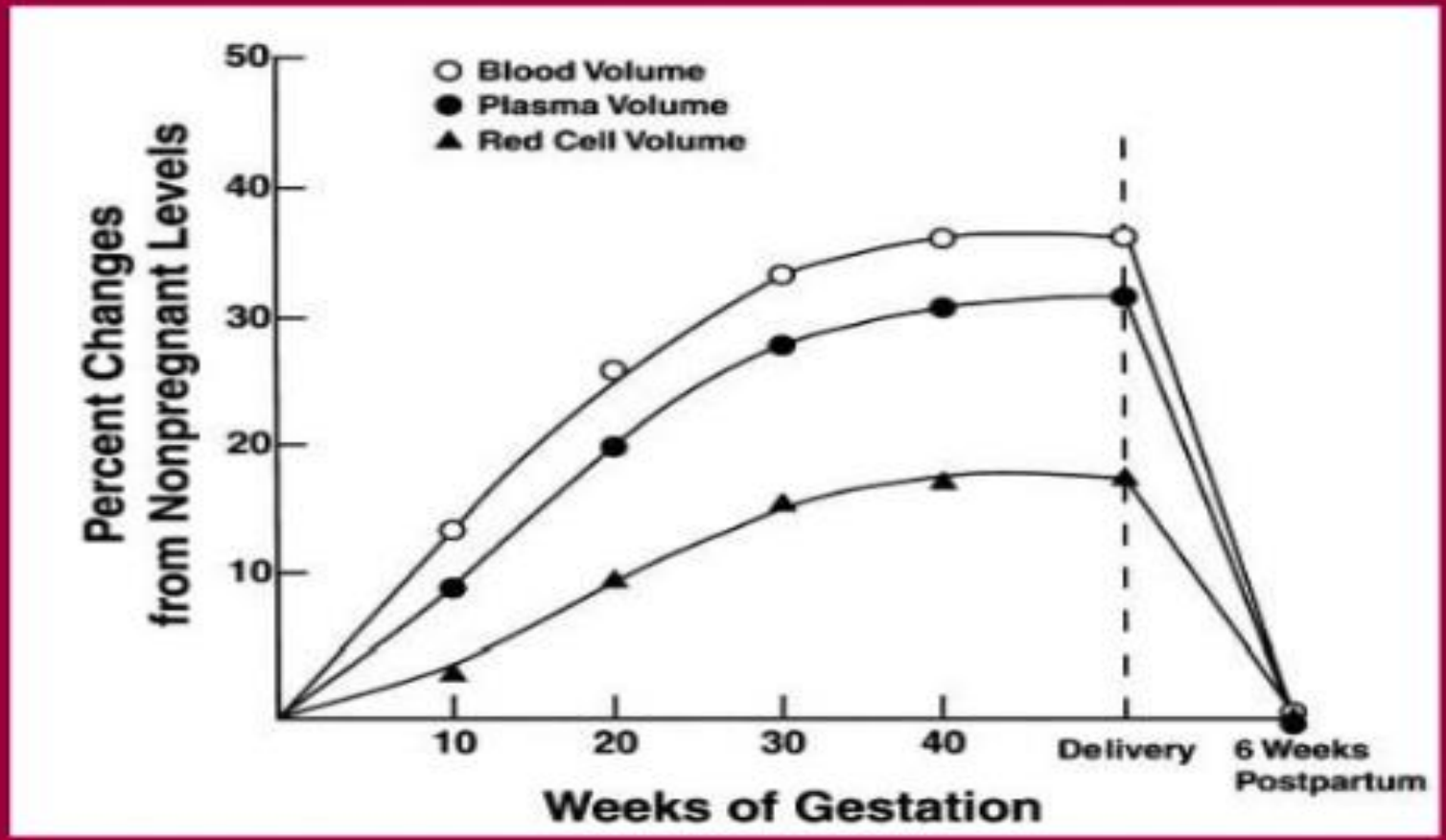
Cardiac Physical Exam

- Physical Exam changes
  - Loud S1
  - Split S1S2
  - Systolic Ejection Murmur
  - Left / upward shift of PMI
- EKG changes
  - Shortened PR intervals
  - Leftward axis deviation
  - Nonspecific ST-wave and T-wave abnormalities

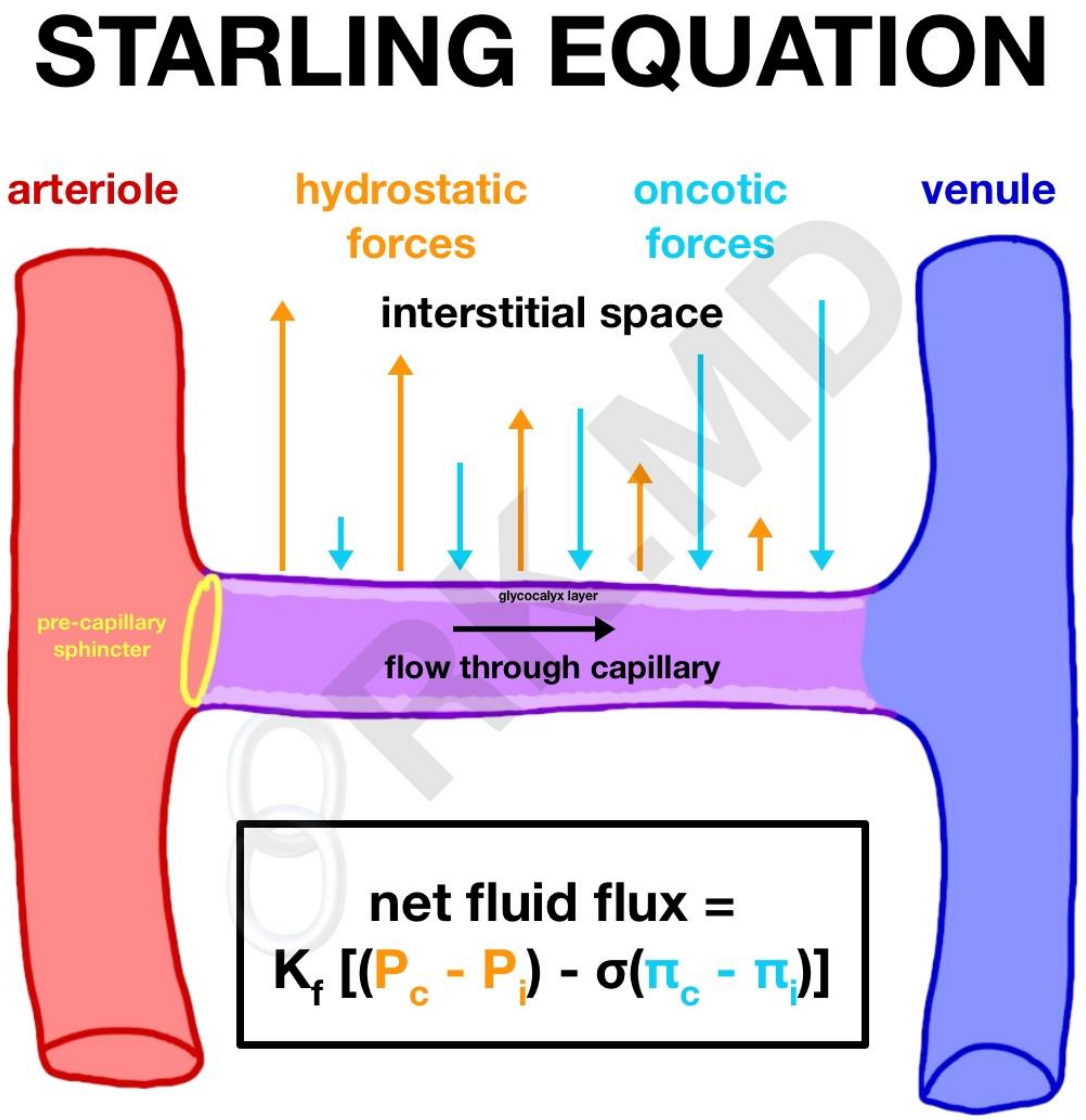
HEMATOLOGIC CHANGES IN PREGNANCY:

A 25-year-old otherwise healthy 4th-year surgical resident comes in to review routine 28w labs drawn last week. You note that the hemoglobin has dropped from 13.2 g/dL, at the initial first trimester visit, to 10.5 g/dL. The resident asks for a reminder of why this has happened. The resident also has some very mild lower extremity edema.

Physiologic anemia in pregnancy



Starling Forces



Platelets and white cells:

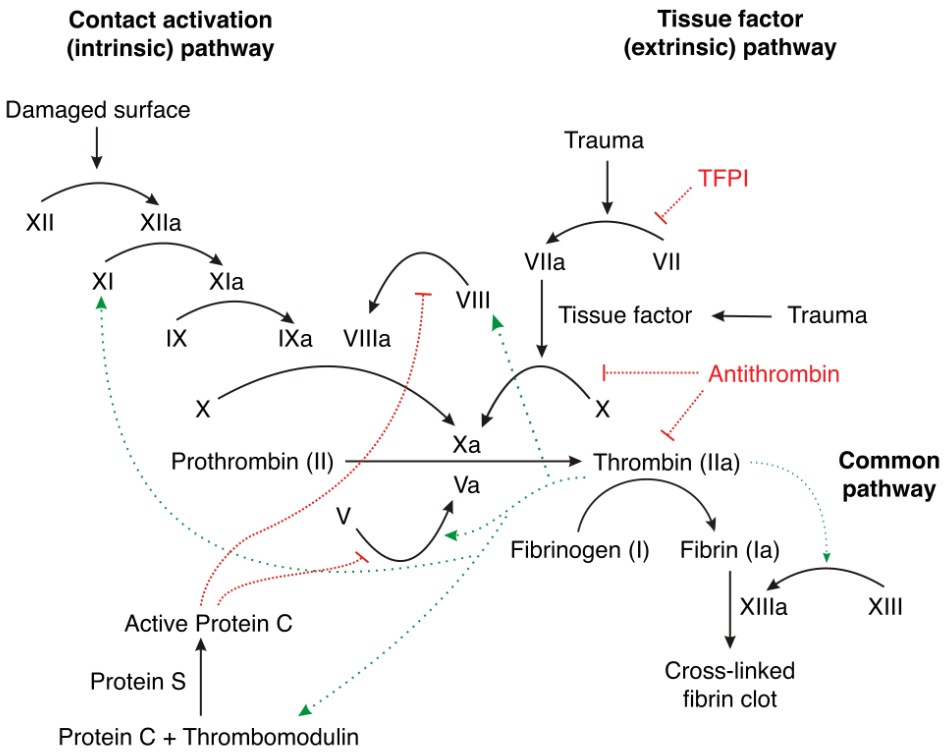
PLATELETS

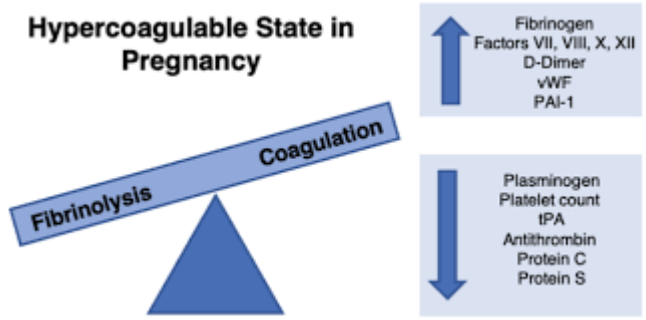
- Stay in the normal range, or are decreased slightly, at the end of pregnancy.
- About 8% of pregnant patients develop gestational thrombocytopenia, with platelet counts between 150,000->70,000.
- The mechanism likely involves an immunologic response, which is very much like mild ITP. Increased platelet consumption may also play a role.

WHITE CELLS

- The white blood cell count rises progressively in pregnancy.
- The average range is about 5->13
- In labor, the white blood cell count can rise precipitously, to the 20-30 range. ***This should not cause alarm unless there are other signs of infection.***

Pregnancy as a Hyper-Coagulable state



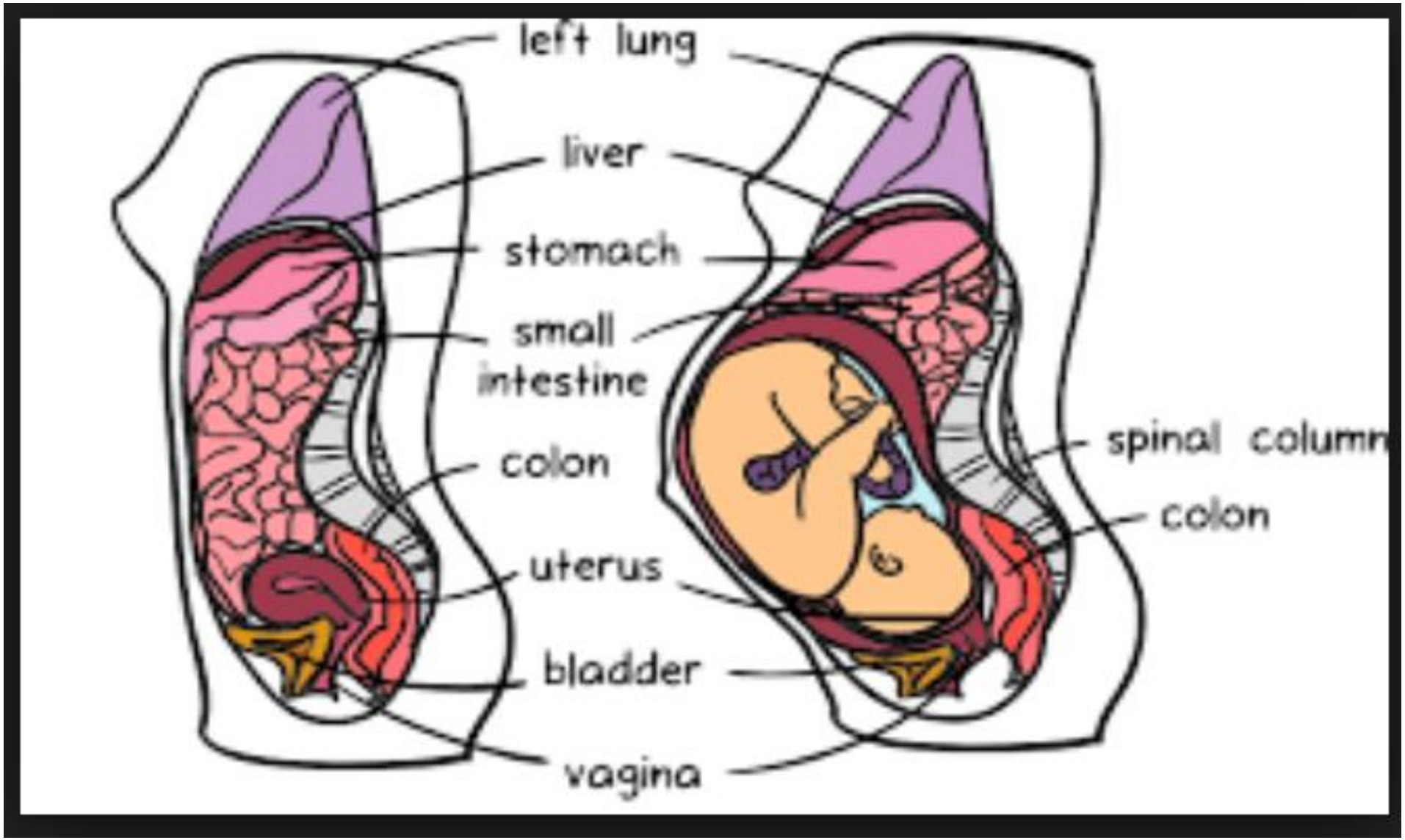


Coagulation factors	Change during pregnancy
Fibrinogen	Increased
VII	Increased
VIII	Increased
X	Increased
XII	Increased
vWF	Increased
Ristocetin cofactor	Increased
Protein S	Decreased
Protein C	No change
Antithrombin III[a]	No change

[a]Decreases during labor and rises later.

PULMONARY CHANGES:

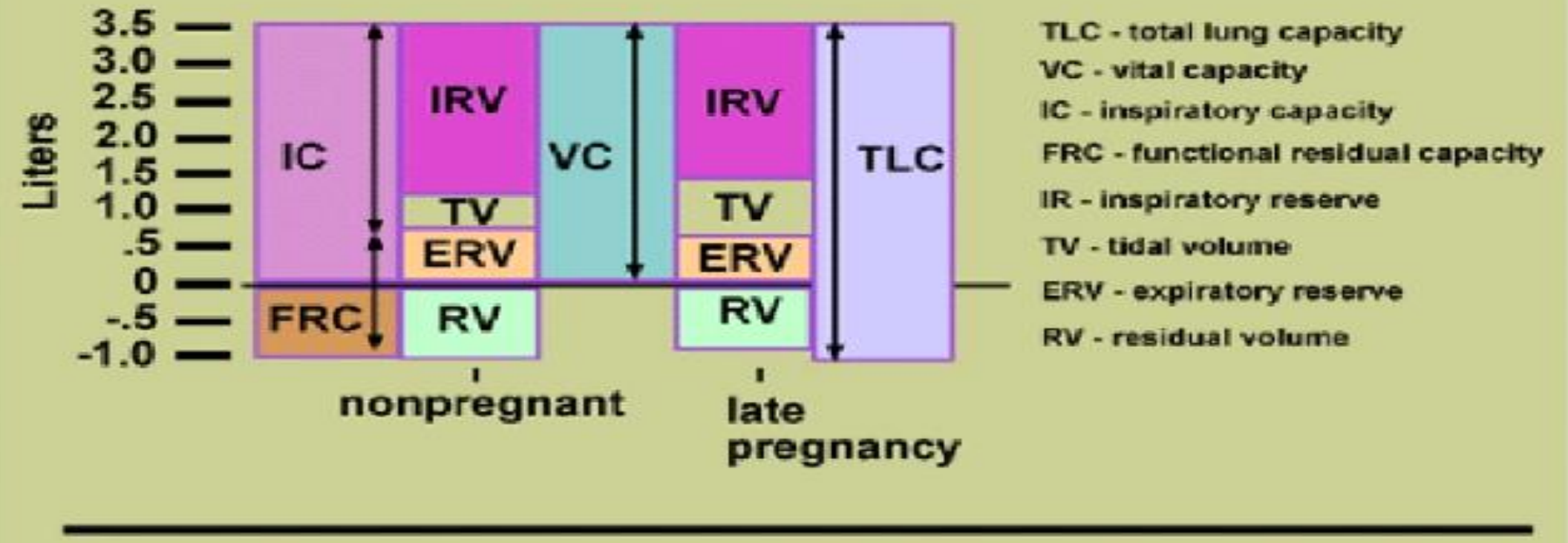
A 26 Y/0 G1P0 patient comes in for a routine office visit at 36 weeks. The patient reports periodic and transient shortness of breath and has difficulty moving air from time to time. The patient does not have a fever, or cough, or any other systemic complaints. Overall, the patient looks and appears perfectly happy and healthy.


left lung
liver
stomach
small intestine
colon
uterus
bladder
vagina
spinal column
colon

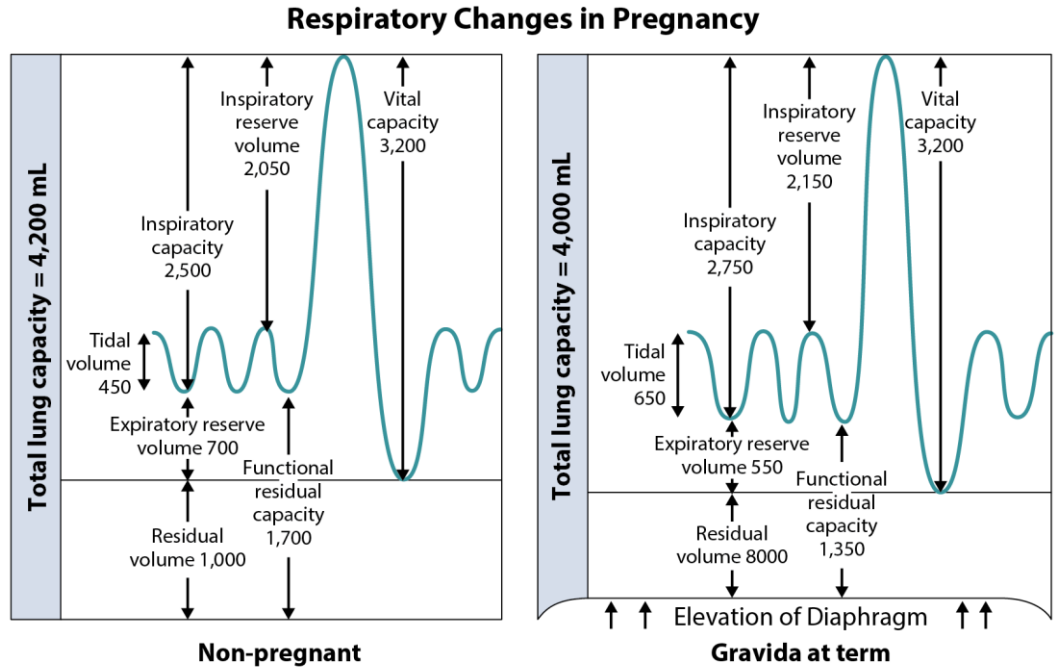
PULMONARY CHANGES: AN EXPLANATION:

- **TIDAL VOLUME (the amount of air we move in and out with every breath cycle): INCREASES BY ABOUT 40% The respiratory rate increases due to progesterone influence and the respiratory drive in the brainstem.**
- These physiologic changes create a mild, chronic hyperventilation which increases alveolar O2.
- Maternal O2 consumption rises 20-40% over nonpregnant levels.
- The increase in O2 consumption results in a net loss of O2 reserve, which can give the sensation of shortness of breath or even tachypnea. This is a major contributor to exercise intolerance, particularly in deconditioned patients.

Lung volumes changes in pregnancy



Respiratory changes in pregnancy graphic:



Effect of Pregnancy on Pulmonary-Function Testing	
• Forced expiratory volume in one second (FEV1)	no change in pregnancy (80–85% of vital capacity)
• Forced vital capacity (FVC)	no change (~3.5 L)
• FEV_1/FVC ratio	no change (>85%)
• Peak expiratory flow rate	no change (~450 L/min)

Blood gas measurement	Non-pregnant adult	Third trimester
pH	7.38–7.44	7.39–7.45
Arterial partial pressure of oxygen (mmHg [kPa])	80–100 (11–13)	92–107 (12.3–14.3)
Arterial partial pressure of carbon dioxide (mmHg [kPa])	35–45 (4.7–5.9)	25–33 (3.3–4.4)
Bicarbonate (mmol/L or mEq/L)	21–30	16–22

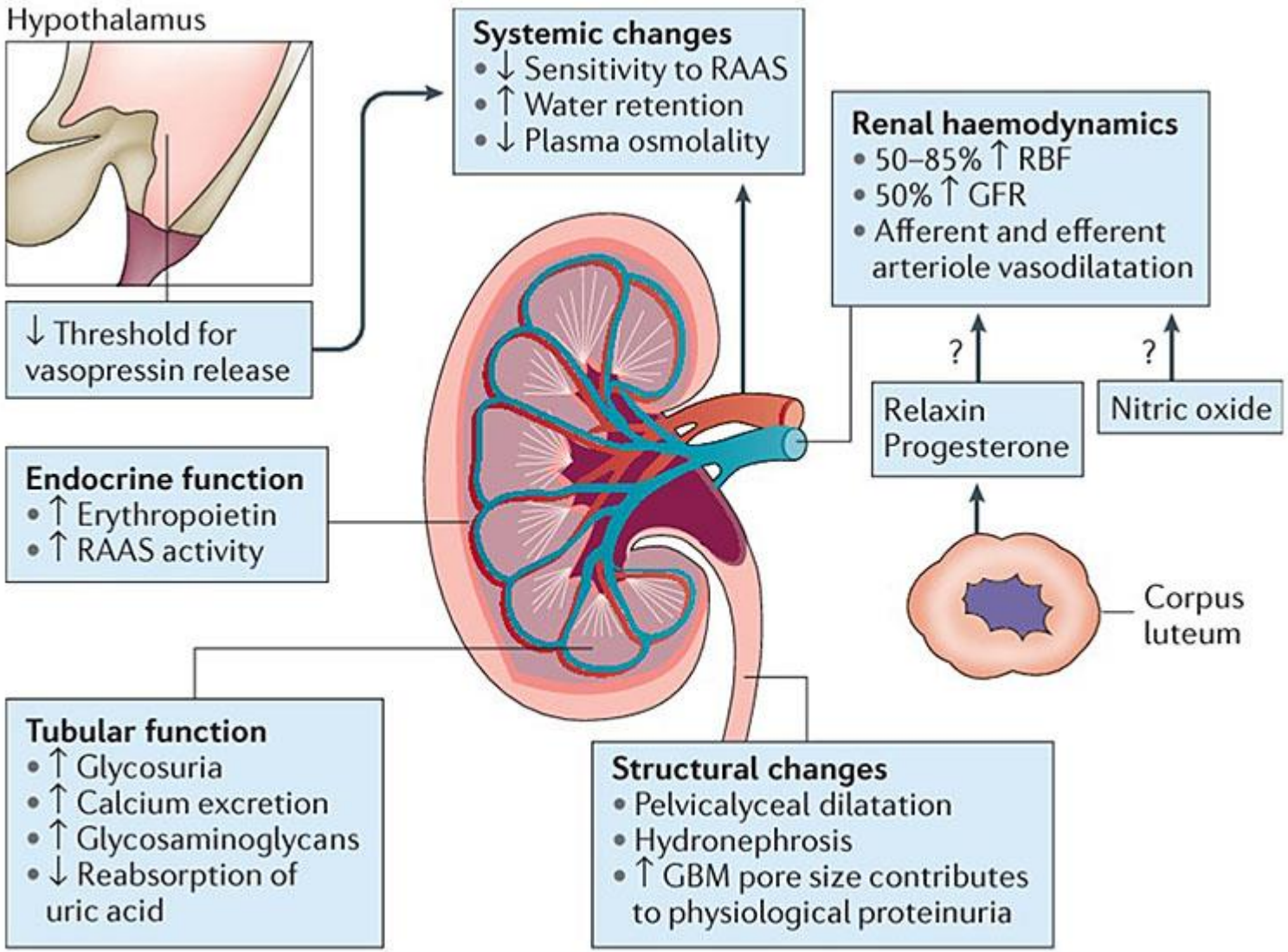
Summary of pulmonary changes:

- **Tidal volume and inspiratory capacity increase.**
- Expiratory reserve volume decreases.
- Minute ventilation increases.
- Respiratory rate increases nominally or stays close to the same.
- **Functional residual capacity decreases.**
- **The overall physiologic changes of pregnancy cause a RESPIRATORY ALKALOSIS.**

RENAL/URINARY CHANGES:

You find yourself at a Labor Day barbecue, and one of your neighbors who is 25 weeks pregnant comes up to you and in the process of discussing youth soccer the patient notes a sense of being thirsty all the time and has a sensation to go to the bathroom a lot. The patient wants to know what you think.

RENAL CHANGES: AN EXPLANATION:



RENAL CHANGES AND TOTAL BODY WATER METABOLISM:

- Total body water increases 6-8 L by the end of pregnancy.
- Pregnancy is a state of chronic volume overload.
- The renin-angiotensin system is entirely upregulated, preserving intravascular volume by increasing sodium reabsorption.
- Sodium metabolism is preserved, secondary to enhanced tubular reabsorption.
- Simultaneously, the heart releases atrial natriuretic peptide, which counteracts the increase in the renin-angiotensin system, and aids in maintaining overall fluid homeostasis.

GASTROINTESTINAL CHANGES IN PREGNANCY:

At a 20-week visit, your patient is delighted to tell you that the daily struggles with morning sickness are over, but that the patient now has fairly severe constipation and reflux. The patient wants to know if this is a trade-off, exchanging one problem for another.

GASTROINTESTINAL CHANGES…AN EXPLANATION OF SEVERAL DIFFERENT PHYSIOLOGIC PHENOMENA:

- Mouth
  - Ptyalism can occur
  - Gums swell and may blee
  - pH unchanged
- Stomach
  - Gastric emptying is NOT delayed
  - GERD rates are elevated – esophageal dysmotility caused by gestational hormones / gastric compression from enlarging uterus/ relaxation of LES.
- Intestines
  - Changes in motility of small inestines / colon are common – increased incidence of constipation. Increased oral-cecal transit times
  - Absorption unchanged but transit time allows for more efficient absorption
- Gallbladder
  - Decreases gallbladder motility
  - Increases lithogenicity of bile
- Liver
  - Liver is no longer palpable due to elevation of diaphragm
  - Most LFTs are unaffected other than alk phosph
  - Pt/ptt are unchanged
- Anus
  - Hemorrhoids more frequent due to increasing pressures / constipation

WEIGHT GAIN IN PREGNANCY

A patient comes into your office who has a normal BMI. The patient wants to know how much weight a person should be expected to gain in a normal pregnancy. The patient states that a concerned friend has constantly been badgering, suggesting that in pregnancy, one needs to “EAT FOR TWO”.

WEIGHT GAIN IN PREGNANCY EXPLAINED:

“Doctor…how much weight should I be gaining?”

• BMI less than 18.5 kg/m²:	28-40 pounds
• Normal BMI 18.5-24.9 kg/m²:	25-35 pounds
• Overweight BMI 25-29.9 kg/m²:	15-25 pounds
• Obese BMI 30-39.9 kg/m²:	11-20 pounds
• Severe obese BMI: >40 kg/m²	0-11 pounds

Calorie Increase in Pregnancy

- 1st Trimester – NONE!
- 2nd Trimester – approx. 350 extra calories
- 3rd trimester – approx. 450 extra calories



Nutrient goals for pregnancy and tolerable upper limits for adult pregnant people

	Source of goal	Nonpregnant/ nonlactating females	Pregnant (age 14 to 18 years)	Pregnant (age 19 to 30 years)	Pregnant (age 31 to 50 years)	Upper limit
Macronutrients						
Protein (as % of kcal)	AMDR	10 to 30%	10 to 30%	10 to 35%	10 to 35%	
Protein	RDA	46 g	71 g	71 g	71 g	
Carbohydrate (as % kcal)	AMDR	45 to 65%	45 to 65%	45 to 65%	45 to 65%	
Carbohydrate	RDA	130 g	175 g	175 g	175 g	
Fiber	14 g per 1000 kcal	25 to 28 g	25 to 34 g	28 to 36 g	25 to 34 g	
Added sugars (as % of kcal)	DGA	<10%	<10%	<10%	<10%	
Total fat (as % of kcal)	AMDR	20 to 35%	25 to 35%	20 to 35%	20 to 35%	
Saturated fat (as % of kcal)	DGA	<10%	<10%	<10%	<10%	
18:3 Linolenic acid	AI	1.1 g	1.4 g	1.4 g	1.4 g	
Minerals						
Calcium	RDA	1000 mg	1300 mg	1000 mg	1300 mg	2500 mg
Iron	RDA	18 mg	27 mg	27 mg	27 mg	45 mg
Magnesium	RDA	310 to 360 mg	400 mg	350 mg	360 mg	350 mg*
Phosphorus	RDA	700 mg	1250 mg	700 mg	700 mg	4000 mg
Zinc	RDA	8 mg	12 mg	11 mg	11 mg	40 mg
Iodine	RDA	150 mcg	220 mcg	220 mcg	220 mcg	1110 mcg
Selenium	RDA	55 mcg	60 mcg	60 mcg	60 mcg	400 mcg
Vitamins						
Vitamin A	RDA	700 mcg RAE	750 mcg RAE	770 mcg RAE	770 mcg RAE	3000 mcg RAE
Vitamin D	RDA	600 international units	600 international units	600 international units	600 international units	4000 international units
Vitamin E	RDA	15 mg	15 mg	15 mg	15 mg	1000 mg
Vitamin K	AI	90 mcg	75 mcg	90 mcg	90 mcg	ND
Vitamin C	RDA	75 mg	80 mg	85 mg	85 mg	2000 mg
Thiamin	RDA	1.1 mg	1.4 mg	1.4 mg	1.4 mg	ND
Riboflavin	RDA	1.1 mg	1.4 mg	1.4 mg	1.4 mg	ND
Niacin	RDA	14 mg	18 mg	18 mg	18 mg	35 mg
Vitamin B6	RDA	1.3 mg	1.9 mg	1.9 mg	1.9 mg	100 mg
Vitamin B12	RDA	2.4 mcg	2.6 mcg	2.6 mcg	2.6 mcg	ND
Choline	AI	425 mg	450 mg	450 mg	450 mg	3500 mg
Folate	RDA	400 mcg DFE	600 mcg DFE	600 mcg DFE	600 mcg DFE	1000 mcg folic acid

In the United States, the reference prepregnancy group weighs 133 pounds (61 kg) at age 19 to 30 years and 125 pounds (57 kg) at age 14 to 18 years. The Dietary Reference Intake Calculator for Healthcare Professionals can be used to estimate calorie needs based on age, sex, height, weight, activity level, and pregnancy or lactation status.

AI: adequate intake (the value that approximates average nutrient intake in a group of healthy individuals; this is used when an RDA cannot be determined); AMDR: acceptable macronutrient distribution range (the range of intake for a macronutrient that is associated with a reduced risk of chronic disease); DFE: dietary folate equivalents (1 mcg DFE = 0.6 mcg folic acid from dietary supplements); DGA: Dietary Guidelines for Americans; ND: not determinable (due to lack of data of adverse effects and concern with regard to lack of ability to handle excess amounts); RAE: retinol activity equivalents; RDA: recommended dietary allowance (the value that meets the requirements of 97% of healthy individuals in a group).

* Applies only to supplemental magnesium.

Adapted from:
1. *US Department of Agriculture and US Department of Health and Human Services. Dietary Guidelines for Americans, 2020-2025, 9th ed. Available at: https://www.dietaryguidelines.gov/ (Accessed on April 11, 2025).*
2. *Institute of Medicine. Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids, The National Academies Press 2000.*
3. *National Institutes of Health. Folate Fact sheet for health professionals. Available at: https://ods.od.nih.gov/factsheets/Folate-HealthProfessional/ (Accessed on April 11, 2025).*

UpToDate®

Micronutrient Requirements

- Ones to focus on
  - Increase protein requirement
  - Increase iron requirement
  - Increase folate requirements
  - Iodine requirements –

ENDOCRINE CHANGES IN PREGNANCY:

At an 8-week initial prenatal visit, a patient wants to talk about thyroid testing which came up on a YouTube feed regarding early pregnancy lab testing. The patient's mother and sister have been treated for hypothyroidism in the past.

ENDOCRINE CHANGES IN PREGNANCY, EXPLAINED:

- Thyroid binding globulin production increases in early pregnancy.
- This allows for an early increase in T3 and T4.
- TSH drops initially, then normalizes. HCG competes for TSH receptor sites, and TSH can be transiently lowered. Nevertheless, these levels largely stay within lab reference ranges.
- fetus cannot synthesize thyroid hormone until after 12 weeks. Fetal thyroid metabolism is entirely dependent on maternal production early on
  - Required for normal neurologic development
  - Cretinism: severe congenital hypothyroidism → short stature, coarse facial features, delayed growth, goiter
- Whether or not to routinely perform screening thyroid levels at the first prenatal visit is controversial. The American Endocrine Society supports this testing, but the American College of Obstetrics and Gynecology does not.

Adrenal Glands

- Pregnancy is a state of increased demand for steroid production
- Increased HPA axis → increased ACTH and cortisol production to support fetus
- Increased RAAS system → increased aldosterone
- Hypercortisolism can lead to maternal fatigue, weight gain, hyperglycemia, edema.
- Size of adrenals does not increase

GLUCOSE METABOLISM IN PREGNANCY:

- Human placental lactogen is produced by the placenta and is only present in pregnancy.
- This hormone decreases maternal insulin sensitivity, which leads to increased blood glucose levels. Glucose is the primary building block for fetal growth.
- Human placental lactogen also contributes to lipolysis, and the release of free fatty acids and cellular glucose stores. Free fatty acids cannot cross the placenta, but the free glucose can.
- This helps support fetal growth even in the face of maternal malnutrition.

MUSKULOSKELETAL CHANGES

You are a third-year Orthopedics resident and are called to the emergency room. The patient you are asked to see is 34 weeks gestation, and has extreme pubic-symphysis pain, and bilateral hip pain.

MUSCULOSKELETAL CHANGES IN PREGNANCY:

- Lordosis of the spine develops, and this phenomenon keeps the center of gravity directly above the patient's legs → causes low back pain

- Widening of the pubic symphysis occurs, from about 3-4 mm up to 8 mm by full term. This can be associated with discomfort and pain.
- Due to the effects of placental relaxin production, as well as the overall effects of progesterone on ligaments and smooth muscles, joint soreness is a common finding.
- High Bone Turnover

Oddballs and curiosities:

- SLEEP: Pregnancy is characterized by sleep disturbances.
  - Fatigue is common, partly due to difficulty becoming comfortable secondary to change in body habitus.
  - There is a noteworthy diminution in REM sleep. This is poorly understood, and likely hormonally mediated.
- SKIN: There are 2 major changes in the skin during pregnancy.
  - A generalized “glow” develops, secondary to increased blood flow.
  - Skin pigment increases, secondary to placental melanocyte stimulating hormone production (melasma or “mask of pregnancy”)
  - Linea nigra is more prominent
  - Spider angiomas, nevi develop
  - Striae develop
- CNS: There are 2 major changes to the CNS system in pregnancy:
  - The pituitary enlarges, as a result of increased blood flow and venous engorgement.
  - Decreased vascular resistance and engorgement in cerebral blood flow can cause headaches. These mostly resolve by the middle of the second trimester, but can persist, particularly in patients who suffer from migraines.
- EYES: Corneal thickeness and intraocular pressures increase. This sometimes results in a change in contact lens and glasses prescriptions following the delivery.
- MICROBIOME: AN EMERGING FIELD:
  - In the vagina, lactobacilli species predominate.
  - Gut bacteria increase.
  - Placental microbiome research is evolving, but it is likely of the maternal-fetal placental interface is not an entirely sterile environment, as was previously thought.

CONCLUSIONS:

- The physiologic changes of pregnancy we reviewed allow the pregnancy to progress and develop normally.
- The physiologic changes of pregnancy happen without harming the fetus or the mother.
- Many of the physiologic changes of pregnancy create signs or symptoms which can be alarming to patients. It is our role as doctors to reassure them when appropriate.
- Perhaps one of the most important jobs of the obstetrician is to be able to determine when a patients concerns reflect a normal physiologic change or may represent a more ominous disease process.

QUESTIONS?